QT40.241, QT40.242

24V, 40A, THREE PHASE INPUT

POWER SUPPLY

- 3AC 380-480V wide-range input
- Three input fuses included
- Version with extra long-lifetime available
- Width only 110mm, weight only 1.5kg
- 95.3% Full load and excellent partial load efficiencies
- 50% BonusPower, 1440W for up to 4s
- 100A Peak current for 10ms for easy fuse tripping
- Active PFC (power factor correction)
- Active filtering of input transients
- Negligible low input inrush current surge
- Full power between -25°C and +60°C
- Current sharing feature for parallel use
- Internal data logging for troubleshooting included
- Remote control of output voltage
- DC-OK relay contact
- Shut-down input
- 3 Year warranty

PRODUCT DESCRIPTION

The most outstanding features of the DIMENSION Q-Series DIN rail power supplies are the extremely high efficiencies and the small sizes, which are achieved by a synchronous rectification and other technological designs.

Large power reserves of 150% support the starting of heavy loads such as DC-motors or capacitive loads. In many cases this allows the use of a unit from a lower wattage class which saves space and money.

High immunity to transients and power surges as well as low electromagnetic emission makes usage in nearly every environment possible.

The integrated output power manager, the three input fuses and near zero input inrush current make installation and usage simple. Diagnostics are easy due to the DC-OK relay, a green DC-OK LED and the red overload LED.

A large international approval package for a variety of applications makes this unit suitable for nearly every application.

SHORT-FORM DATA

Output voltage	DC 24V	nominal
Adjustment range	24 - 28V	
Output current	40 – 34.3A	continuous
	60 – 51.5A	short term (4s)
Output power	960W	continuous
	1440W	short term (4s)
Output ripple	< 100mVpp	20Hz to 20MHz
Input voltage	3AC 380-480V	-15%/+10%
	3AC 480V	+20% max.5s
Mains frequency	50-60Hz	±6%
AC Input current	1.58 / 1.3A	at 3x400 / 480Vac
Power factor	0.92 / 0.93	at 3x400 / 480Vac
AC Inrush current	typ. 4.5A peak	
Efficiency	95.3 / 95.2%	at 3x400 / 480Vac
Losses	47.3 / 48.4W	at 3x400 / 480Vac
Temperature range	-25°C to +70°C	operational
Derating	24W/°C	+60 to +70°C
Hold-up time	typ. 25 / 25ms	at 3x400 / 480Vac
Dimensions (wxhxd)	110x124x127mm	
Weight	1500g / 3.3lb	

ORDER NUMBERS

Power Supply	**QT40.241**	24-28V Standard unit
	QT40.242	24-28V Standard unit extra long-lifetime
Accessory	ZM2.WALL	Wall mount bracket
	UF20.241	Buffer unit
	YR80.241	Redundancy module

MAIN APPROVALS

For details and the complete approval list, see chapter 22.

UL 508

UL 60950-1

Marine

DNV.COM/AF

Marine

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TERMINOLOGY AND ABREVIATIONS

PE and ⏚ symbol	PE is the abbreviation for **P**rotective **E**arth and has the same meaning as the symbol ⏚.
Earth, Ground	This document uses the term “earth” which is the same as the U.S. term “ground”.
t.b.d.	To be defined, value or description will follow later.
AC 400V	A figure displayed with the AC or DC before the value represents a nominal voltage with standard tolerances (usually ±15%) included. E.g.: DC 12V describes a 12V battery disregarding whether it is full (13.7V) or flat (10V)
400Vac	A figure with the unit (Vac) at the end is a momentary figure without any additional tolerances included.
50Hz vs. 60Hz	As long as not otherwise stated, AC 230V parameters are valid at 50Hz mains frequency.
may	A key word indicate flexibility of choice with no implied preference.
shall	A key word indicate a mandatory requirement.
should	A key word indicate flexibility of choice with a strongly preferred implementation.

1. INTENDED USE

This device is designed for installation in an enclosure and is intended for commercial use, such as in industrial control, process control, monitoring, measurement equipment or the like.
Do not use this device in equipment, where malfunctioning may cause severe personal injury or threaten human life without additional appropriate safety devices, that are suited for the end application.
If this device is used in a manner outside of its specification, the protection provided by the device may be impaired.

2. INSTALLATION INSTRUCTIONS

WARNING Risk of electrical shock, fire, personal injury or death.
- Turn power off before working on the device. Protect against inadvertent re-powering.
- Do not open, modify or repair the device.
- Use caution to prevent any foreign objects from entering the housing.
- Do not use in wet locations or in areas where moisture or condensation can be expected.
- Do not touch during power-on and immediately after power-off. Hot surfaces may cause burns.

Obey the following installation instructions:
This device may only be installed and put into operation by qualified personnel.
This device does not contain serviceable parts. The tripping of an internal fuse is caused by an internal defect.
If damage or malfunction should occur during installation or operation, immediately turn power off and send unit to the factory for inspection
Install device in an enclosure providing protection against electrical, mechanical and fire hazards.
Install the device onto a DIN rail according to EN 60715 with the input terminals on the bottom of the device. Other mounting orientations require a reduction in output current.
Make sure that the wiring is correct by following all local and national codes. Use appropriate copper cables that are designed for a minimum operating temperature of 60°C for ambient temperatures up to +45°C, 75°C for ambient temperatures up to +60°C and 90°C for ambient temperatures up to +70°C.
Ensure that all strands of a stranded wire enter the terminal connection. Use ferrules for wires on the input terminals. Unused screw terminals should be securely tightened.
The device is designed for pollution degree 2 areas in controlled environments. No condensation or frost is allowed.
The enclosure of the device provides a degree of protection of IP20. The enclosure does not provide protection against spilled liquids.
The device is designed for overvoltage category II zones. Below 2000m altitude the device is tested for impulse withstand voltages up to 4kV, which corresponds to OVC III according to IEC 60664-1.
The device is designed as “Class of Protection I” equipment according to IEC 61140. Do not use without a proper PE (Protective Earth) connection.
The device is suitable to be supplied from TN, TT or IT mains networks, grounding of one phase is allowed. The continuous voltage between the input terminals and the PE potential must not exceed 528Vac.
A disconnecting means shall be provided for the input of the device.
The device is designed for convection cooling and does not require an external fan. Do not obstruct airflow and do not cover ventilation grid!
The device is designed for altitudes up to 5000m (16400ft). Above 2000m (6560ft) a reduction in output current is required.
Keep the following minimum installation clearances: 40mm on top, 20mm on the bottom, 5mm left and right side. Increase the 5mm to 15mm in case the adjacent device is a heat source. When the device is permanently loaded with less than 50%, the 5mm can be reduced to zero.
The device is designed, tested and approved for branch circuits up to 32A (IEC) and 30A (UL) without additional protection device. If an external fuse is utilized, do not use circuit breakers smaller than 6A B- or C-Characteristic to avoid a nuisance tripping of the circuit breaker.
The maximum surrounding air temperature is +70°C (+158°F). The operational temperature is the same as the ambient or surrounding air temperature and is defined 2cm below the device.
The device is designed to operate in areas between 5% and 95% relative humidity.

3. AC-INPUT

AC input	nom.	3AC 380-480V	suitable for TN, TT and IT mains networks, grounding of one phase is allowed
AC input range		3x 323-528Vac	continuous operation
		3x 528-576Vac	short-time up to 5s
Input frequency	nom.	50–60Hz	±6%
Turn-on voltage	typ.	3x 305Vac	steady-state value, load independent, see Fig. 3-1
Shut-down voltage	typ.	3x 275Vac	steady-state value, load independent, see Fig. 3-1

		3AC 400V	3AC 480V	
Input current	typ.	1.58A	1.3A	at 24V, 40A, symmetrical phase voltages, see Fig. 3-3
Power factor*)	typ.	0.92	0.93	at 24V, 40A, see Fig. 3-4
Start-up delay	typ.	500ms	600ms	see Fig. 3-2
Rise time	typ.	35ms	35ms	at 24V, 40A, resistive load, 0mF see Fig. 3-2
	typ.	40ms	40ms	at 24V, 40A, resistive load, 40mF see Fig. 3-2
Turn-on overshoot	max.	500mV	500mV	see Fig. 3-2

*) The power factor is the ratio of the true (or real) power to the apparent power in an AC circuit

Fig. 3-1 **Input voltage range**

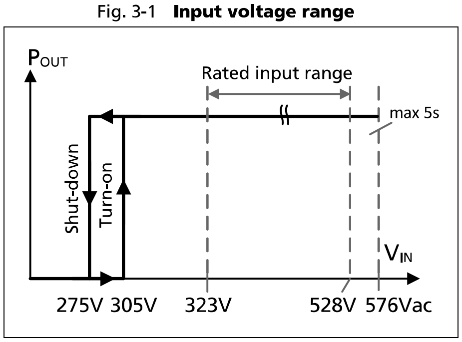


Fig. 3-2 **Turn-on behaviour, definitions**

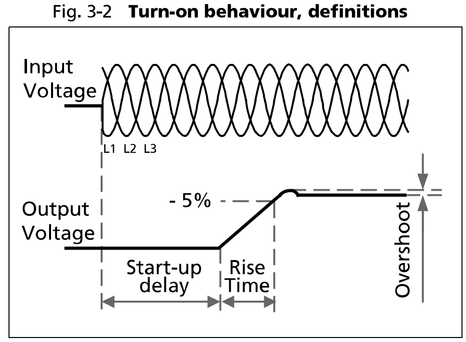


Fig. 3-3 **Input current vs. output load at 24V**

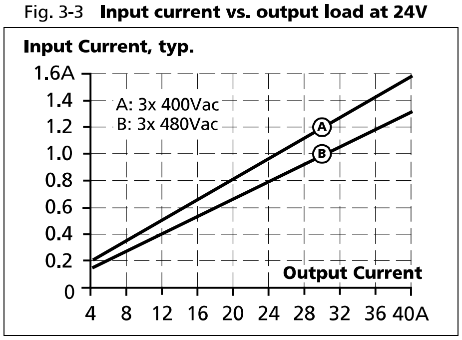
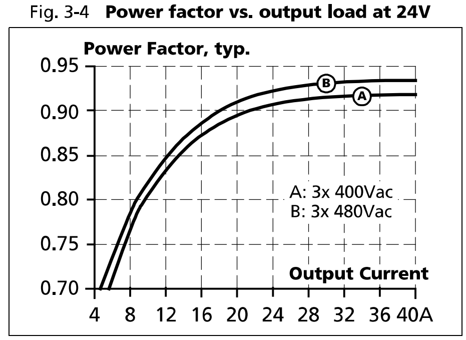


Fig. 3-4 **Power factor vs. output load at 24V**



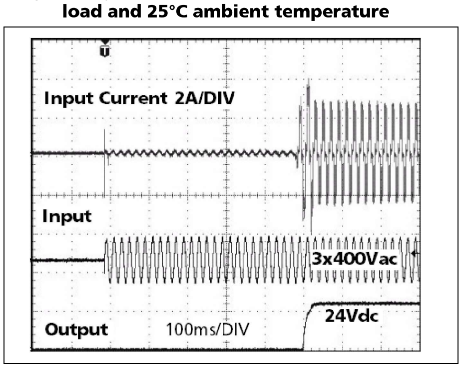
4. INPUT INRUSH CURRENT

The power supply is equipped with an active inrush current limitation circuit, which limits the input inrush current after turn-on to a negligible low value. The input current is usually smaller than the steady state input current.

		3AC 400V	3AC 480V	
Inrush current*)	max.	$6A_{peak}$	$6A_{peak}$	over entire temperature range
	typ.	$4.5A_{peak}$	$4.5A_{peak}$	over entire temperature range
Inrush energy	max.	$1.5A^2s$	$1.5A^2s$	over entire temperature range
Inrush delay	typ.	500ms	600ms	

*) The charging current into EMI suppression capacitors is disregarded in the first microseconds after switch-on.

Fig. 4-1 **Typical turn-on behaviour at nominal load and 25°C ambient temperature**



5. DC-INPUT

Do not operate this power supply with DC-input voltage.

6. OUTPUT

Output voltage	nom.	24V	
Adjustment range		24-28V	guaranteed
	max.	30V***)	at clockwise end position of potentiometer
Factory setting	typ.	24.1V	±0.2%, at full load, cold unit, in "single use" mode
	typ.	24.1V	±0.2%, at full load, cold unit, in "parallel use" mode
	typ.	25.1V	at no load, cold unit, in "parallel use" mode
Line regulation	max.	10mV	3x323-576Vac
Load regulation	max.	50mV	in "single use" mode: static value, 0A→40A, see Fig. 6-1
	typ.	1000mV	in "parallel use" mode: static value, 0A→40A, see Fig. 6-2
Ripple and noise voltage	max.	100mVpp	20Hz to 20MHz, 50Ohm
Output current	nom.	40A	continuously available at 24V, see Fig. 6-1 and Fig. 6-2
	nom.	34.3A	continuously available at 28V, see Fig. 6-1 and Fig. 6-2
	nom.	60A	short term (4s) available BonusPower®*), at 24V, see Fig. 6-1, Fig. 6-2 and Fig. 6-4
	nom.	51.5A	short term (4s) available BonusPower®*), at 28V, see Fig. 6-1, Fig. 6-2 and Fig. 6-4
	typ.	100A	up to 10ms, output voltage stays above 20V, see Fig. 6-4, This peak current is available once every second. See chapter 26.2 for more peak current measurements.
Output power	nom.	960W	continuously available at 24-28V
	nom.	1440W*)	short term available BonusPower®*) at 24-28V
BonusPower® time	typ.	4s	duration until the output voltage dips, see Fig. 6-3
BonusPower® recovery time	typ.	7s	overload free time to reset power manager, see Fig. 6-5
Overload behaviour		cont. current	see Fig. 6-1
Short-circuit current**)	min.	40A	continuous, load impedance 25mOhm, see Fig. 6-1
	max.	44A	continuous, load impedance 25mOhm, see Fig. 6-1
	min.	60A	short-term (4s), load impedance 25mOhm, see Fig. 6-1
	max.	68A	short-term (4s), load impedance 25mOhm, see Fig. 6-1
	typ.	46A	continuous, load impedance <10mOhm
	max.	51A	continuous, load impedance <10mOhm
	typ.	190A	up to 10ms, load impedance <10mOhm, see Fig. 6-4
Output capacitance	typ.	10 200µF	included in the power supply

*) **BonusPower®, short term power capability (up to typ. 4s)**
The power supply is designed to support loads with a higher short-term power requirement without damage or shutdown. The short-term duration is hardware controlled by an output power manager. This BonusPower® is repeatedly available. Detailed information can be found in chapter 26.1. If the power supply is loaded longer with the BonusPower® than shown in the bonus-time diagram (see Fig. 6-3), the max. output power is automatically reduced to 960W.

**) Discharge current of output capacitors is not included.

***) This is the maximum output voltage which can occur at the clockwise end position of the potentiometer due to tolerances. It is not guaranteed value which can be achieved. The typical value is about 28.5V.

Fig. 6-1 **Output voltage vs. output current in "single use" mode, typ.**

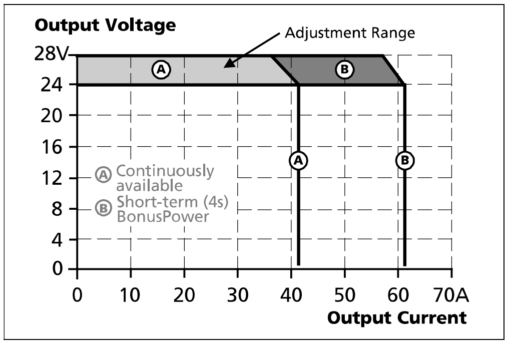


Fig. 6-2 **Output voltage vs. output current in "parallel use" mode, typ.**

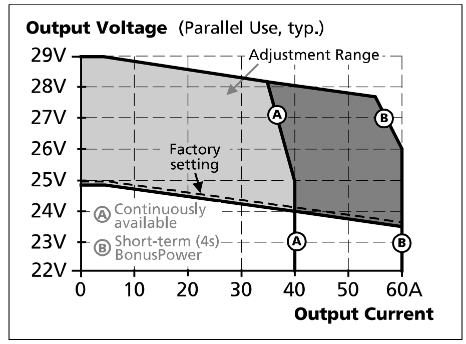


Fig. 6-3 **Bonus time vs. output power**

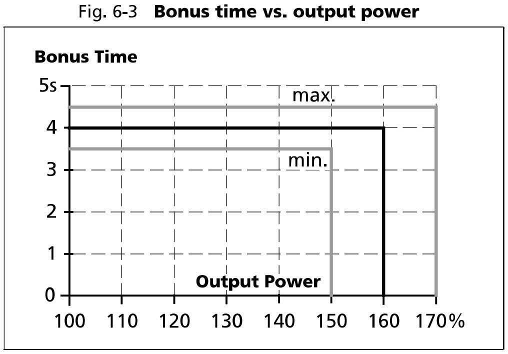


Fig. 6-4 **Dynamic overcurrent capability, typ.**

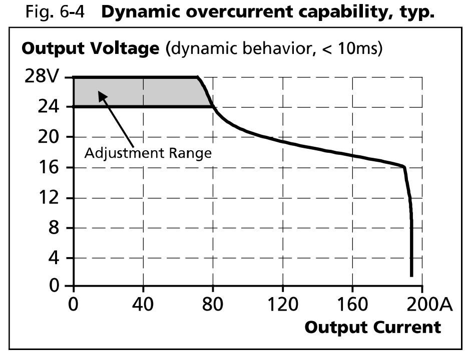
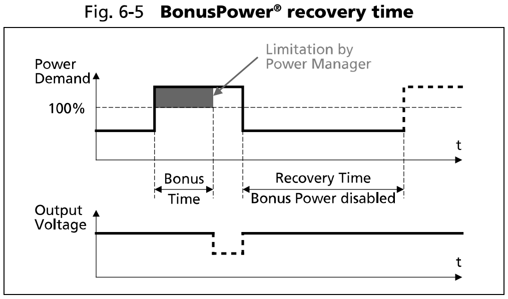


Fig. 6-5 **BonusPower® recovery time**



The BonusPower® is available as soon as power comes on and after the end of an output short circuit or output overload.

Fig. 6-6 **BonusPower® after input turn-on**

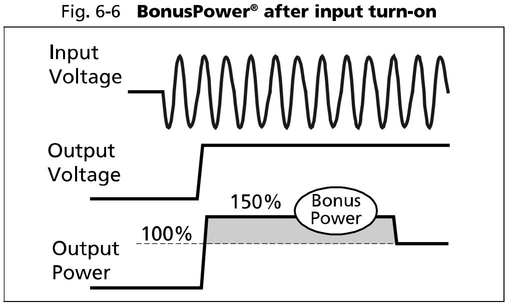
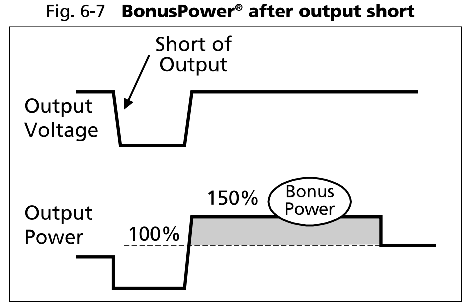


Fig. 6-7 **BonusPower® after output short**



7. HOLD-UP TIME

		3AC 400V*)	3AC 480V*)	
Hold-up Time	typ.	50ms	50ms	at 24V, 20A, see Fig. 7-1
	min.	40ms	40ms	at 24V, 20A, see Fig. 7-1
	typ.	25ms	25ms	at 24V, 40A, see Fig. 7-1
	min.	20ms	20ms	at 24V, 40A, see Fig. 7-1

*) Curves and figures for operation on only 2 legs of a 3-phase system can be found in chapter 26.4

Fig. 7-1 **Hold-up time vs. input voltage**

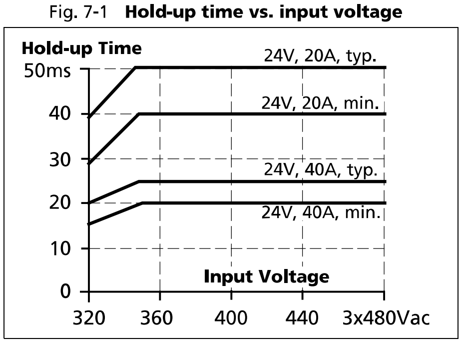


Fig. 7-2 **Shut-down behaviour, definitions**